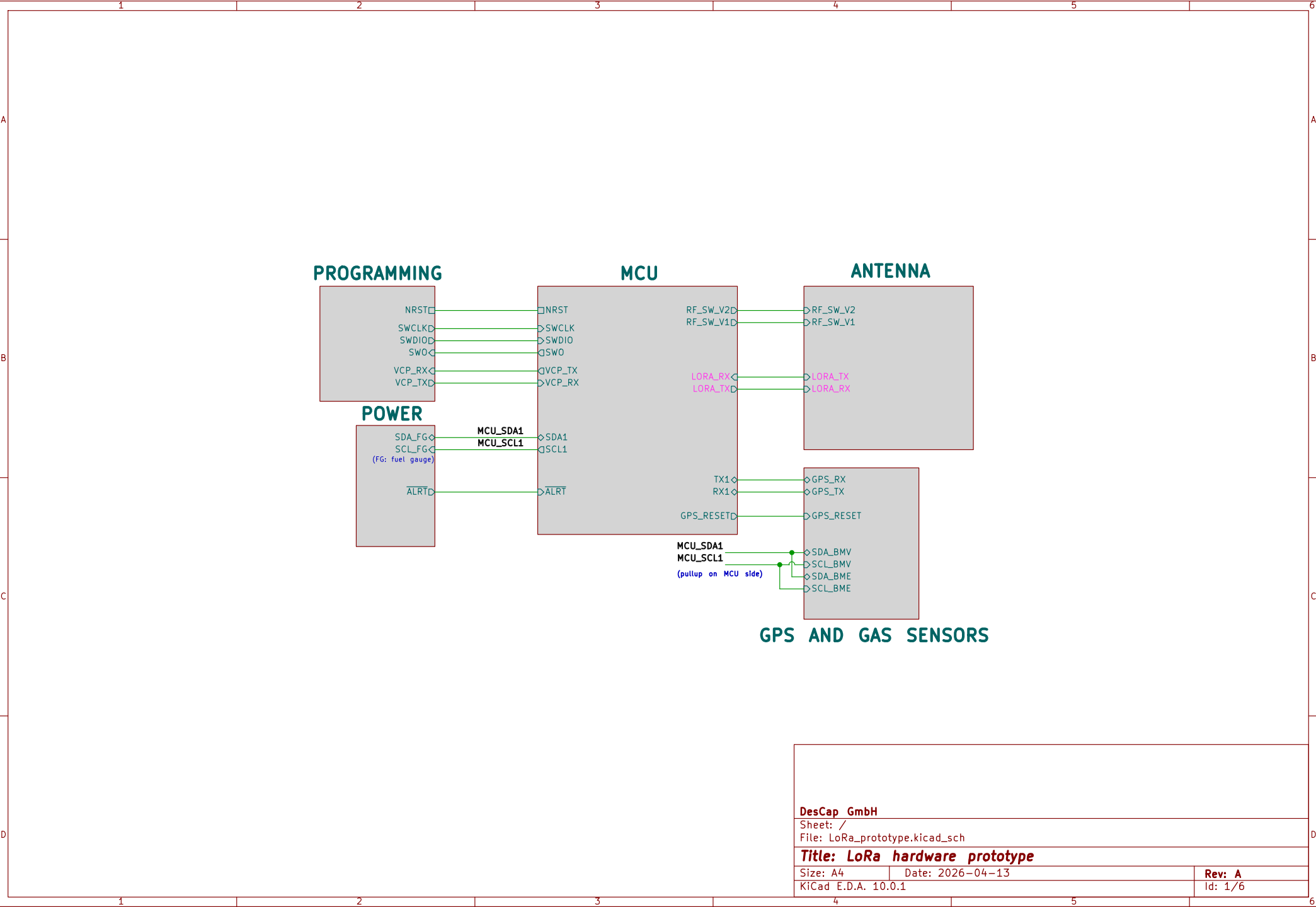


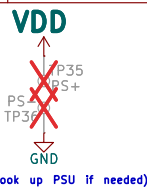
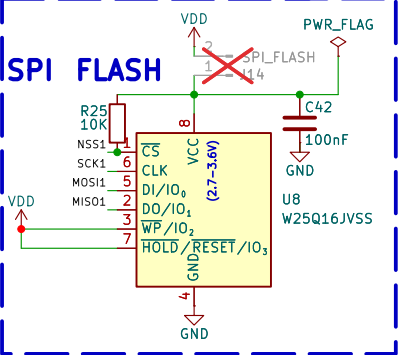
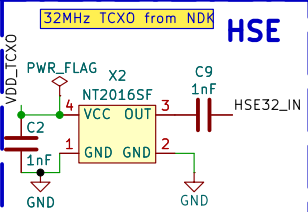
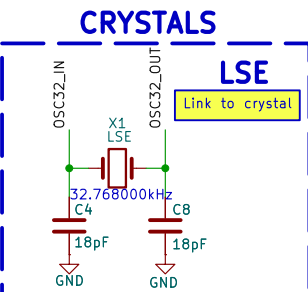
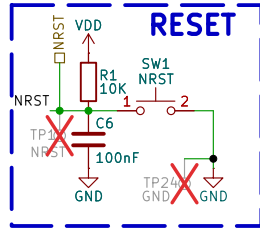
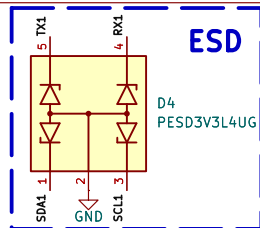


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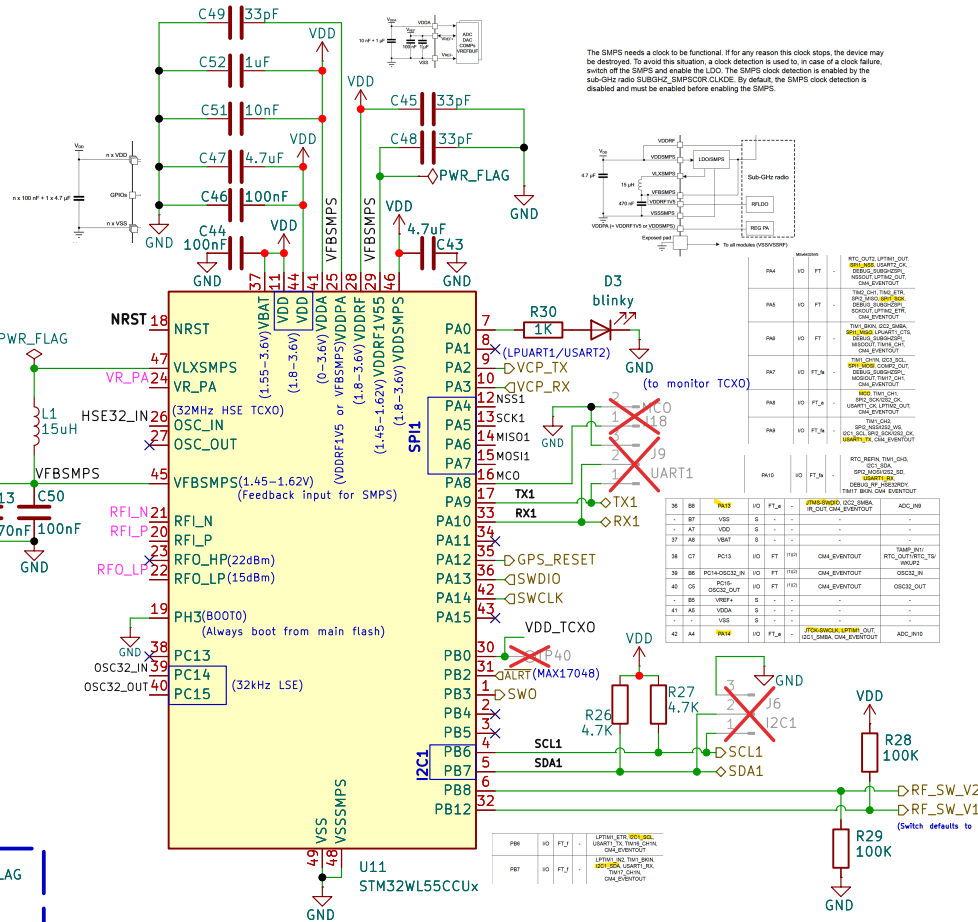
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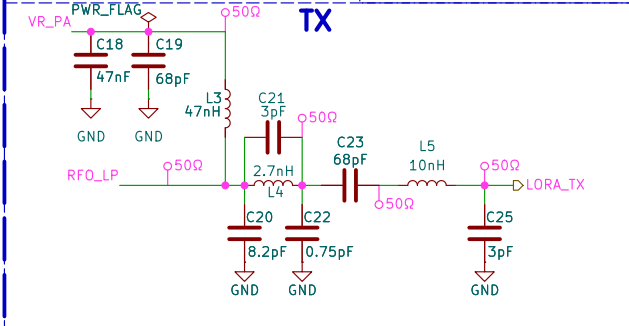
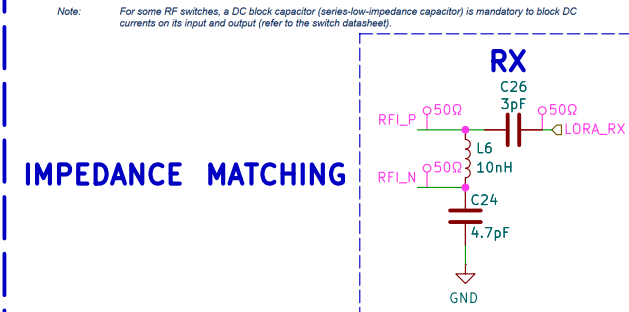
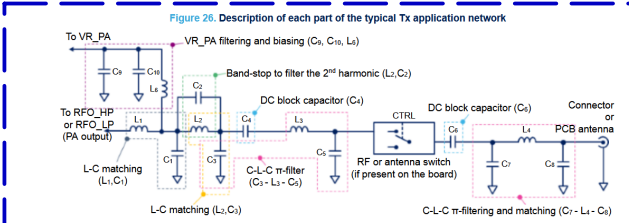
Size: A4	Date: 2026-04-13	Rev: A
KiCad E.D.A. 10.0.1		Id: 1/6

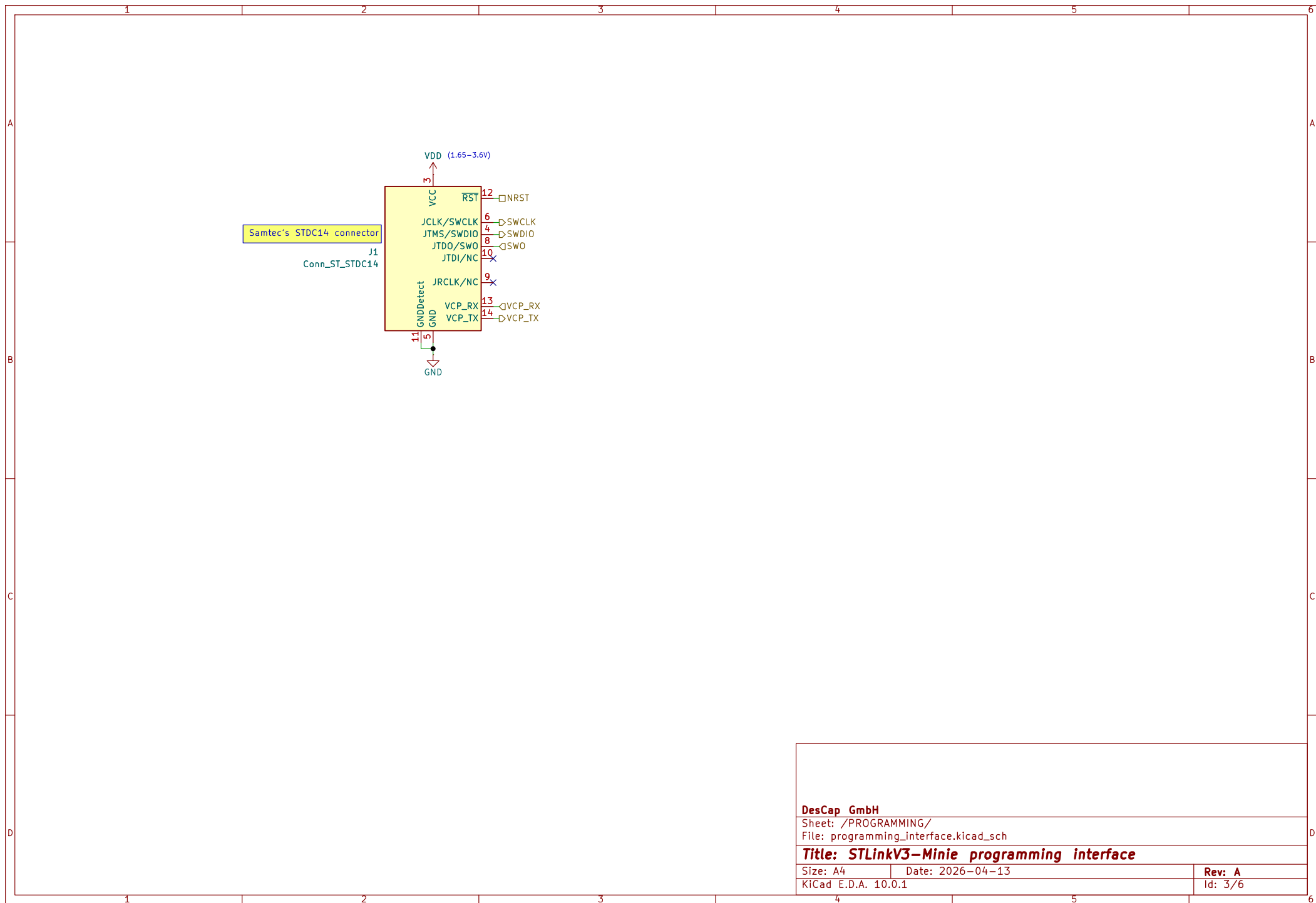


This design is based on the reference design MB1720



RF components:
High-Q Murata GJM15 series for caps (0.1–33pF with 0402 footprints)
High-Q Murata LQW15AN series for inductors (1.3–75nH 0402 footprints)





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Sheet: /PROGRAMMING/
File: programming_interface.kicad_sch

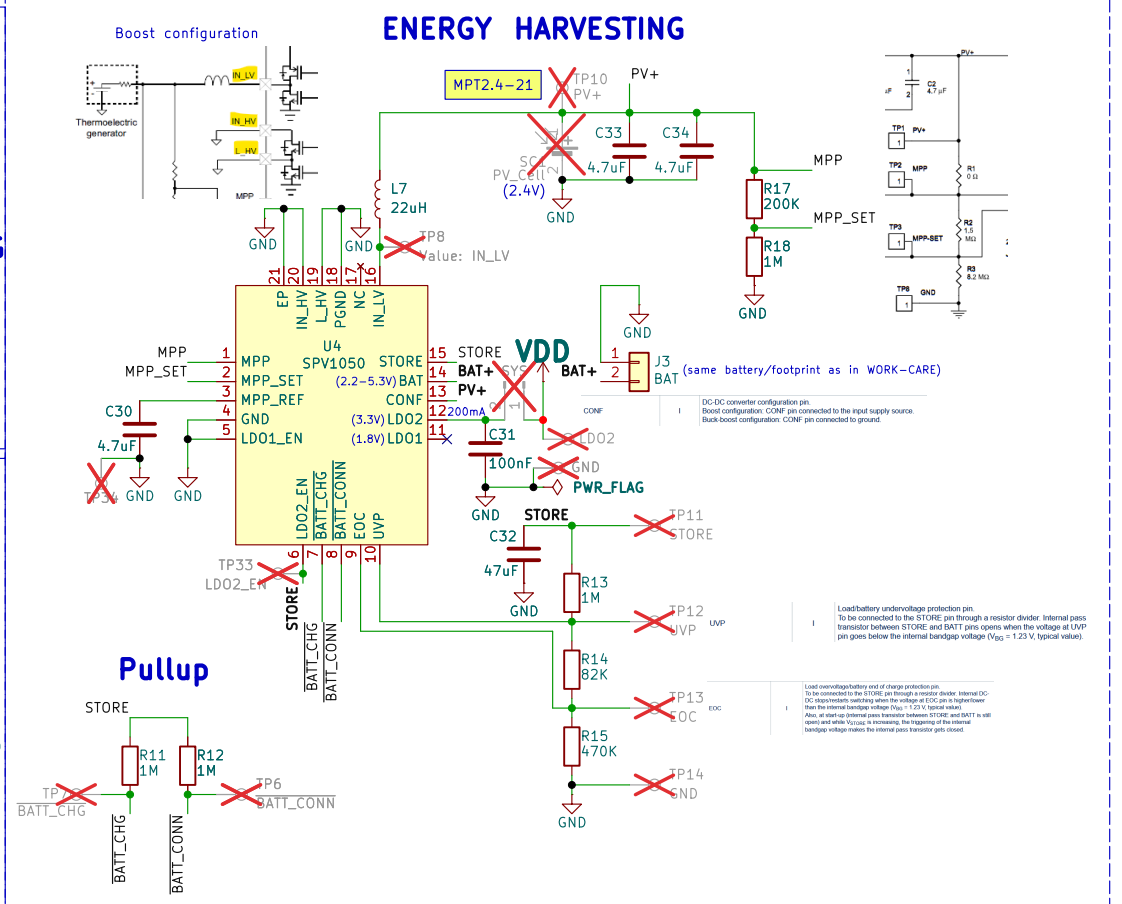
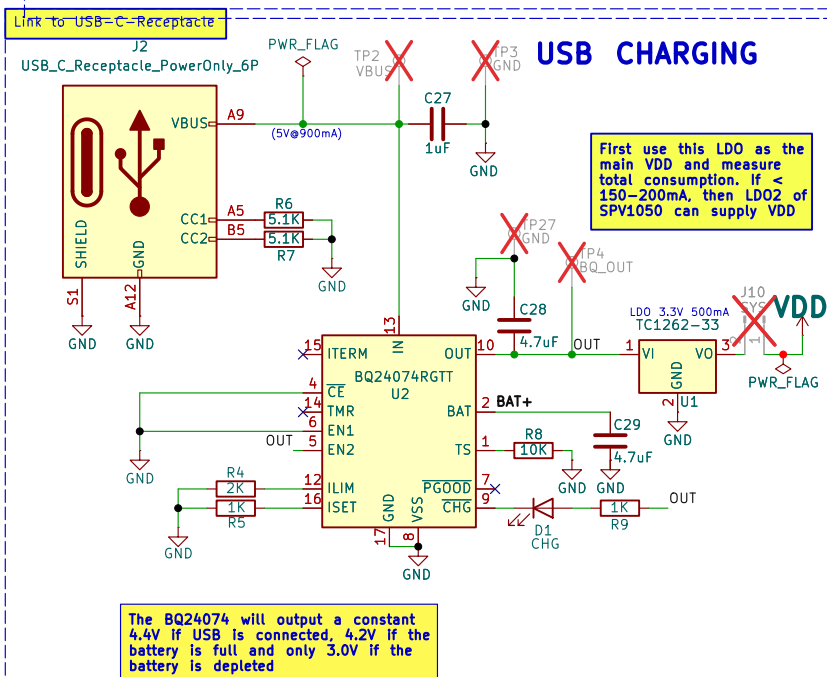
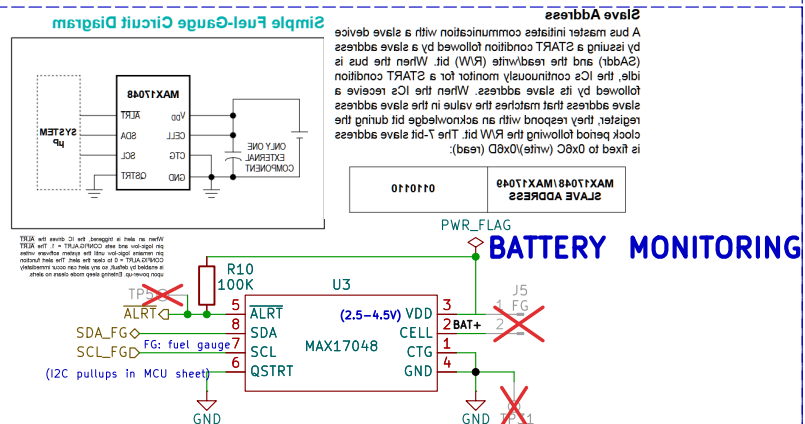
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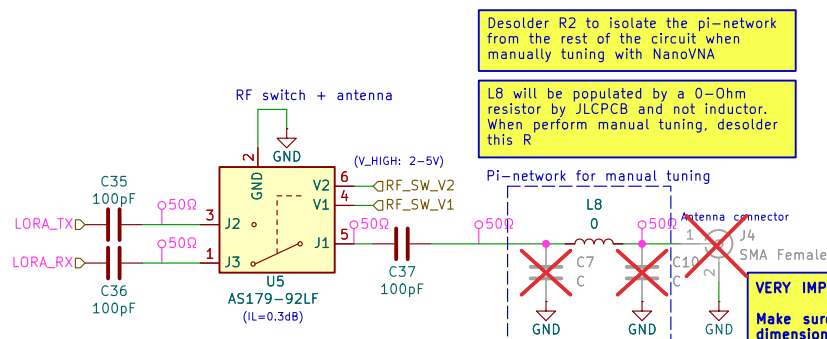
Size: A4 Date: 2026-04-13

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Rev: A

Id: 3/6





Overview SMA edge launch connector for antenna for 1.6mm-thick 4-layer PCBs

Part	Price in \$	Stock	Legs distance in mm
SMA-J-P-X-ST-EM1	4.48	201	1.73
SMA-KE-347-H13.5-1.6	0.50	2269	1.75
BWSMA-KE-P001	0.42	37404	1.60
SMA-KE-P903	0.99	2097	1.75
SMA-KE-P902	0.66	490	1.73

VERY IMPORTANT:

Make sure the SMA connector has the same dimension has PCB thickness (1.6mm for a 4-layer board)

Samtec's SMA-J-P-H-ST-EM1 connector for 1.57mm board thickness

NOTES:

To further reduce cost, consider using SMA connectors from HJ, which are readily available on JLCPCB

Table 4. Truth Table (V_{DD} = 2 to 5 V)¹

V1	V2	J1-J2	J1-J3
V _{DD}	0	Isolation	Insertion loss
0	V _{DD}	Insertion loss	Isolation

¹ Any state other than described in this table places the device in an undefined state. An undefined state does not damage the device.

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Sheet: /ANTENNA/

File: antenna.kicad_sch

Title:

Size: A4

Date: 2026-04-13

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Id: 5/6

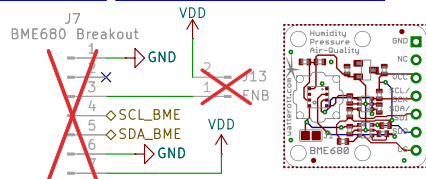
GAS SENSORS

The 7-bit device address is 111011x. The 6 MSB bits are fixed. The last bit is changeable by SDO value and can be changed during operation. Connecting SDO to GND results in slave address 1110110 (0x76); connection it to V_{DDIO} results in slave address 1110111 (0x77), which is the same as BMP280's I²C address. The SDO pin cannot be left floating; if left floating, the I²C address will be undefined.

CSB=VDDIO	I ² C
CSB=GND	SPI

Link to Watterott BME680 Breakout Board

BME680



I²C address: 0x76

The protocol select (PS) pin is a logic input to configure the type serial interface, SPI or I²C. Figure 25 shows how the PS pin can be connected to select the SPI or the I²C protocol. The SPI protocol is selected when this pin is connected to a logic low (VSSD). If this pin is connected to a logic high (VDDIO), the I²C protocol is selected. If this pin is not connected (not recommended), the I²C protocol is selected. The pin state is latched during power-up by the digital core.

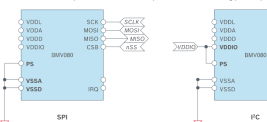
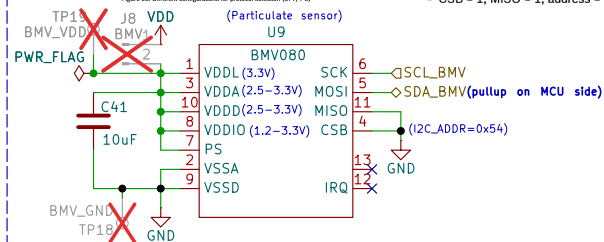


Figure 25: Different configurations for protocol selection (SPI/I²C)



Below are some other I²C address selection possibilities:

- CSB = 0, MISO = 0, address = 0x54
- CSB = 0, MISO = 1, address = 0x55
- CSB = 1, MISO = 0, address = 0x56
- CSB = 1, MISO = 1, address = 0x57

GPS MODULE

EX_ANT 11 I External active antenna RF input

TIMER	7	O	An open drain output signal can be used to control GPS module main power on/off.	V _{DDmin} = 0.3 V V _{DDmax} = 0.4 V V _{DDmin} = 1.1 V V _{DDmax} = 3.1 V	It belongs to RTC domain. Functioned, keep this pin open or connect to GND externally.
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AADET_N	8	IO	Active antenna detection	V _{DDmin} = 0.2 V V _{DDmin} = 1.2 V	Functioned, keep this pin open. Refer to Chapter 4.3
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If unused, keep this pin open.

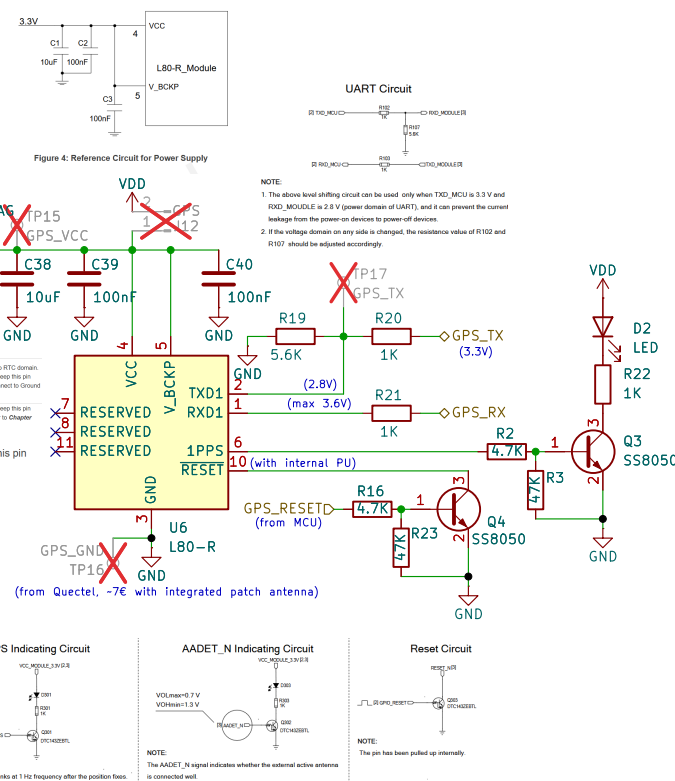


Figure 4: Reference Circuit for Power Supply

NOTE:
1. The above level shifting circuit can be used only when TXD_MCU is 3.3 V and RXD_MODULE is 2.8 V (power domain of UART), and it can prevent the current leakage from the power-on devices to power-off devices.
2. If the voltage domain on any side is changed, the resistance value of R102 and R107 should be adjusted accordingly.

1PPS Indicating Circuit

AADET_N Indicating Circuit

Reset Circuit

NOTE:
The 1PPS indicator blinks at 1 Hz frequency after the position fixes.

NOTE:
The AADET_N signal indicates whether the external active antenna is connected well.

NOTE:
The pin has been pulled up internally.

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Sheet: /GPS AND GAS SENSORS/

File: gps.kicad_sch

Title: GPS and gas sensor modules

Size: A4

Date: 2026-04-13

Rev: A

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Id: 6/6